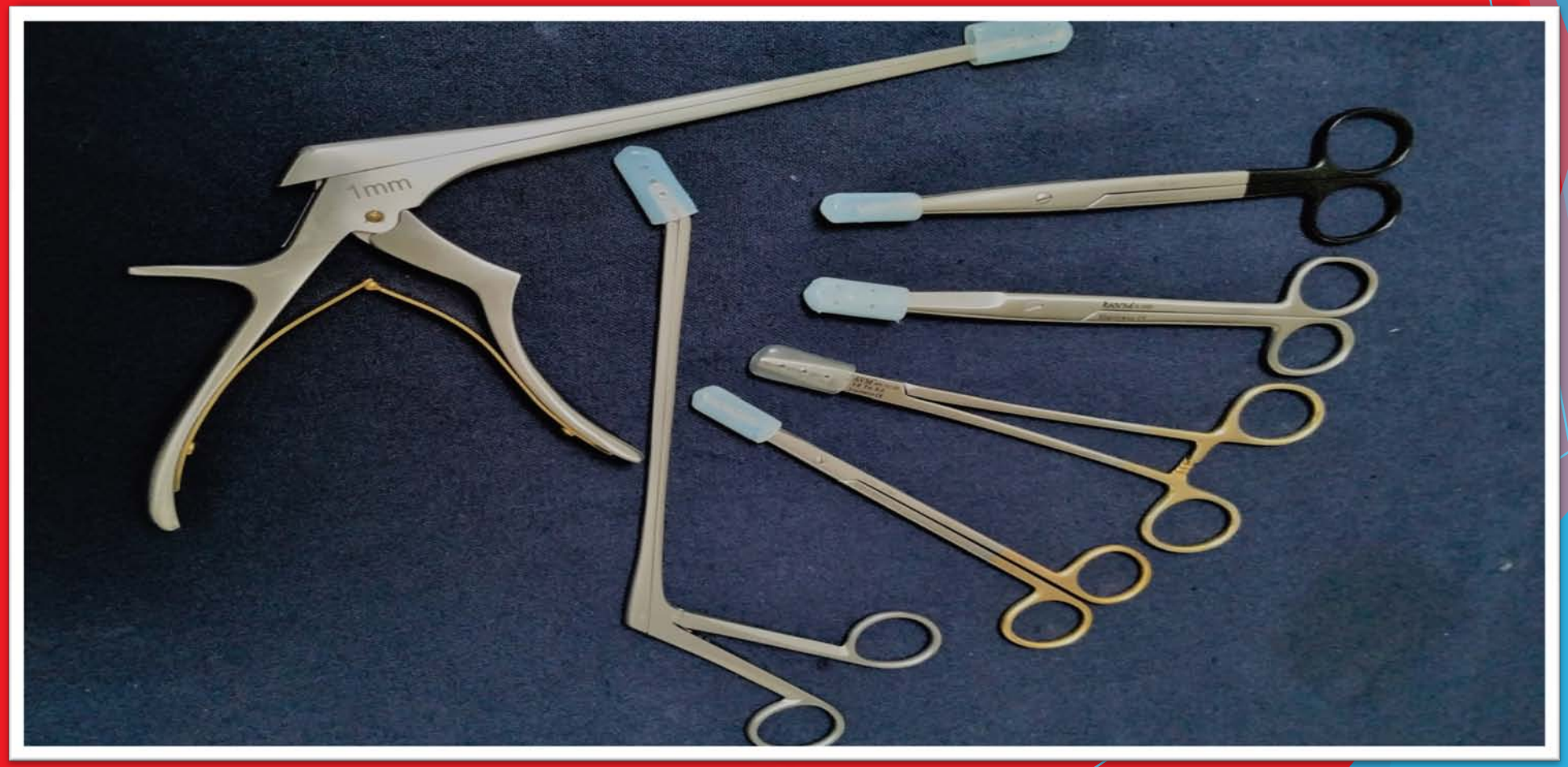
# Care and maintenance of surgical instruments



# About the instruments

AVM Surgical instruments are designed to perform many specific functions. The use of any instrument for tasks other than those for which they are intended may result in damage or breakage. The proper cleaning, handling and sterilization procedures will ensure that your surgical instrument will perform as intended and extend its useful life.

## Cleaning :

**step 1.Maintain moisture** :Immediately after instrument use ,place the instruments in an instrument tray/container and cover with a towel moistened with sterile distilled water.

**Step 2.Enzymatic soak**: Disassemble instruments as appropriate. Immerse instruments in an approved enzymatic solution per enzyme manufacturer's recommendations with instrument jaws in open position.

**Step 3.Rinse** :Remove from enzymatic soak after the time period recommended by the enzymatic manufacturer and rinse with tap water distil water.

**Step 4.Clean instruments**: Using a small, clean, hand-held brush, clean the handle and ratchet mechanism while immersed in the cleaning solution. Clean the shank or the shaft of the instrument by wiping with a soft cloth such as a disposable lab sponge. For those instruments with an accessible channel with a soft bristled ,clean brush. Remove the soil from the jaws, tips, box lock, and /or hinge mechanism with a small soft bristled, clean brush.

**Step5. Rinse**: Rinse instruments by immersing in tap water and wiping with clean, soft cloth

**Step 6. Ultrasonic Cleaning***:* Place instruments in a mesh bottom stainless steel instrument basket. Place basket in sonic cleaner. Follow the recommendation of the sonic cleaner manufacturer as to cycle times, cleaning solution ,suspension of the basket (e.g., basket should not sit on bottom of sonic cleaner), conditioning of the water, etc.

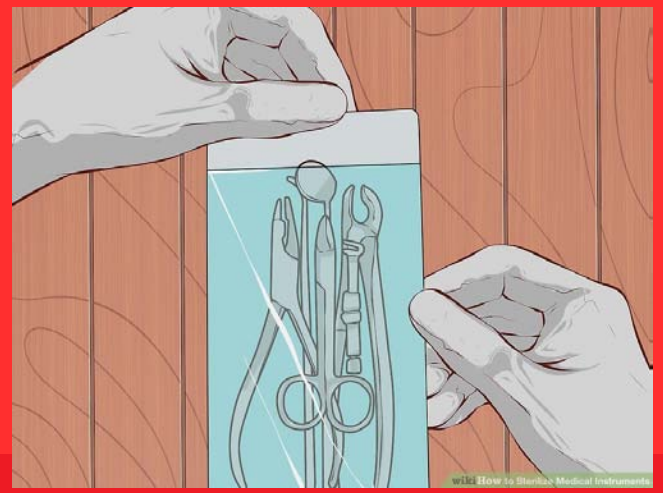
**Step.7.Visual inspection :**Visually inspect the instrument for cleanliness.

***Step 8.* Drying:** Before instruments are wrapped for storage they must be thoroughly dry. Remaining moisture may result in corrosion that will weaken the instruments and lead to breakage during use.

***Step 9.* lubrication** : Before instruments are prepared for sterilization, the use of water-solution instruments lubricant are recommended. This will help keep cleaning joints moving freely and will aid in protecting the entire surface from mineral deposition. Note the ultra sonic cleaners remove all the lubrication therefore this maintenance procedure should be done routinely after ultrasonic cleaning and before sterilization.

# Sterilization

- Sterilization is only as effective as the cleaning process prior to sterilization. Other factors impacting the effectiveness of sterilization include the types of microorganisms on the device, the number and location of microorganisms, and the type and amount of organic material surrounding the microorganism that may protect it from the steam or chemical sterility (biofilm). Additionally, the effectiveness of sterilization depends upon key parameters of the process, which, depending on the process used, include temperature, humidity, pH, water quality, contact time, pressure, and chemical concentration.



**Proper Use:** Instruments should be used during surgery exclusively for their intended purpose. Misusing instrument will lead to damage that is usually not repairable; for example, a hemostat that is used to clamp tubing can become misaligned and quickly break.

**Protect Instruments:**

The use of neutral pH detergents is vital to the maintenance of surgical instruments. Contact with acidic or alkaline solution will remove the instruments' protective barrier of chromium oxide, often leading to corrosion, pitting and breakage.

**Certain compounds:** are highly corrosive to stainless steel and will cause serious damage despite the passivated protective surface. If instruments are inadvertently exposed to any of the following substances, they should be rinsed immediately.

**Instruments should never be exposed to:**

Aqua regia
Ferric chloride
Hydrochloric acid
iodine
sulfuric acid

The following substances should be avoided whenever possible:

Aluminum chloride
Mercury chloride
Barium chloride
Potassium permanganate
Bichloride of mercury
Potassium thiocyanate
Calcium chloride
Saline
Carbolic acid
Sodium hypochlorite
Chlorinated lime
Stannous chloride
Dakin's solution

The use of a water soluble instrument lubricant is an integral step in maintaining the long-life of the surgical instrument. Lubrication will prevent the friction of metal on metal and preserve the smooth function of the instrument thus avoiding corrosion by friction. Furthermore, constant use of lubricating agents prevents the “sticking” of hinged parts and will aid in protecting . the entire instrument surface from mineral deposits. Non-water-soluble lubricants cannot be sterilized and, therefore, should never be used.

During manual cleaning, never use steel wool, wire brushes or highly abrasive detergent cleansers. These will damage the instrument’ protective surface and lead to corrosion.

Any kind of corrosion will lead to rust on steel.
Because rust particles can be transferred from one instrument to another during disinfection, cleaning or sterilizing, corroding instruments must be separated to prevent the formation of rust on other instrument.

Due to the heating up and cooling down cycles of the sterilization process, a surgical instrument with a closed ratchet may suffer from tension stress which causes stress cracks in the joints or a deterioration of the clamping force. Therefore, such instruments must be sterilized either in an open position or closed only to the first tooth of the ratchet.

Every effort should be made to protect sharp cutting edges and fine working tips during all maintenance procedures. Avoiding loading retractors and other heavy items on top of delicate and hollow instruments.

## Cleaning:

Before first use, all instruments must be thoroughly cleaned prior to sterilization.

It is important to rinse instruments that have been exposed to blood and saline solution before these substances dry. Blood and saline solution are highly corrosive. In addition, blood will also cause a stain that is difficult to remove. Instruments should, therefore, be kept moist Immediately following use, either by placing in an appropriate basin of sterile distilled water or covered with a moist towel.

**Drying**:

Before instruments are wrapped for storage, they must be thoroughly dry. Remaining moisture, particularly in box locks and hinged may result in corrosion that will weaken the instrument and lead to breakage during use.

If instruments are to be stored after sterilization, they have to be stored dry until used again. Excessive condensation maximum weight for loaded perforated trays. Drying is facilitated by wrapping the perforated trays with cloth within the container or external paper packing.

**Inspection**: Inspection is a vital part of proper care and maintenance. Instruments in need of repair will not perform accurately in surgery and breakage is likely to occur. worn ratchets, loose box locks and misaligned jaws can be repaired at a fraction of the cost of new instruments. Contact your local AVM representative for information regarding a cost-effective instrument repair program.

When cleaning, sterilizing and rinsing instruments, it is important to use solution with a pH as near 7.0 (neutral) as possible. Distilled water is recommended because it is free of the many compounds which exist is ordinary water. These substances, alone, cause stains and when tap water is combined with some detergents it will from insoluble deposits on the instruments. Manual cleaning will be most effective with a solution of luke-warm, distilled water and a neutral pH detergent. This should be followed by a two stage rinse, first warm then cool.

The box lock, Cavity of any Forceps and hinge portion of the instrument must be kept clean and free of debris. A build-up of substance in this area will cause instruments to become stiff and eventually break. manual cleaning should remove all visible residue. It is essential to keep the box lock and hinged open during any automated cleaning procedure.

· Follow the manufacturer's specifications when using automatic washer-sterilizers. Use a free-rinsing, low sudsing detergent with a neutral pH (7.0) and do not alter prescribed cycle times. A high-sudsing detergent may clean effectively but will often leave residual deposits on the instruments.

**Diagnosing Spots and Stains:** It is common for instruments to become stained or spotted despite the best efforts of the manufacturers and the hospital staff. In nearly all cases these problems are the result of minerals deposited upon the surfaces of the instruments. Adhering to proper technique during cleaning and sterilizing procedures will prevent most staining occurrences. However, they will sometimes arise very suddenly and will not disappear on their own. The following identifies the various instrument-related problems hospitals may encounter.

# 1. Problem: Light or dark water.

Spots are often the result of slow evaporation of water condensation on the instruments. Often, the mineral deposit, which is related to the mineral content of the water, remains on the instrument. Using distilled water for sterilizing procedures will help to eliminate the formation of these residual spots. It is also important to follow manufacturer's specific autoclave operating instructions. Spots may result, for example, if the autoclave doors are opened before the steam is completely vented. An additional cause of spotting can be traced to the instrument wraps. During laundering procedures it is vital that detergents are thoroughly rinse is prepared so that the wraps have a pH between 6.8 and 7.0. This will inhibit spotting during steam sterilization.

**Cause:** Mineral content in tap water or condensate in the sterilizer caused by inadequate drying.

**solution:** Ensure that water is purified through reverse osmosis and deionization, each of which removes the necessary contaminants.

## 2. Problem: Rust-colored surface corrosion

It is very unlikely for surgical grade steel to rust. What appears to be rust is often residual organic matter in box locks or mineral deposits which have been baked onto the surface of the instrument. In localities where the water has a high iron content, for example, an iron deposit will result in a metallic film on the instrument. This is not the fault of the instrument and can be prevented with the use of distilled or demineralized water during cleaning procedures

**Cause:** Iron content, chloride content, or softening agents in steam pipes.

**Solution**: Clean steam lines in adherence to sterilizer manufacturer IFUs.

**3.Problem:brown stain:** Detergents containing polyphosphates may dissolve copper elements in the sterilizer. This result in copper being deposited on the instruments by an electrolytic reaction. The hospital may try a different detergent or check the quantities used. Usually a dull blue or brown stain is simply a build-up of oxidation on the surface. This film is harmless and will actually protect the instrument from serious corrosion.

**Cause:** Polyphosphate cleaning compounds that are incompatible with the local water supply, leaving a chromic oxide film on the instruments.

**Solution:** Ensure that water is purified through reverse osmosis and deionization.

**4.Problem: Blue Stain**: Blue Stain are usually the result of cold sterilization techniques. It is important to prepare the solution according to exact proportions and to change the solution when recommended. Serious corrosion may occur if the solution is used beyond the manufacturer’s specified time limit. The use of distilled water and a rust inhibitor in the solution will help retard discoloration.

**Cause:** Certain chemical sterilizing or disinfecting solutions.

**Solution:** Verify that the right type and amount of sterilizing and disinfecting solutions keep being used.

**5. Problem:** Purplish black stain.

**Cause:** The presence of chlorides, or detergents that contain ammonia or from amines in the steam lines.

**Solution:** Clean steam lines in adherence to sterilizer manufacturer IFUs.

**6. Problem:** Rust colored deposits

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**Cause:** Inadequate cleaning or drying, inappropriate water used for rinsing, agents that are not
thoroughly rinsed off or electrolyte deposits
from exposed metal underneath chipped chrome plating on stainless steel, presence of chlorides.

**Solution:** Ensure autoclave trays are being loaded correctly to affirm adequate cleaning
and drying. If due to electrolyte deposits, repair and replace chrome plating.

**NEEDLE HOLDER**: The use of suture needle larger than those recommended may result in damage and invalidate your warranty. Please see the suture size guide for AVM needle holder for recommendation regarding the use of suture needle with AVM instrument a copy of this publication is packaged with every AVM needle holder and also available upon request.

**SCISSORS:** Do not cut suture gauze or wire with scissors unless it was designed to cut those materials. All AVM super cut & ultra sharp scissors must be returned to AVM for complete refurbishment and sharpening in order to maintain the unique cutting edge.

**AVM PREMIUM SCISSORS: Sharpness of AVM scissors is guaranteed for 12 months. If needed, reshaping will be performed at no charge during the 12 months period.**

**TITANIUM INSTRUMENTS: Different shades of blue may found with the extensive line of anodized AVM titanium instrumentation. This is not a defect, but a phenomenon which occur due to slight variation of manufacture technique or alloy composition. This is a cosmetic characteristic only and does not compromise the high quality of your AVM instruments.**

# PROPER STORAGE OF THE INSTRUMENTS

# AVM HEALTHCARE

Warrantees the Quality of All it's Products
Any item which found to be defective,
Because of material or poor
craftsmanship, will be Repaired or
Replace Free of charges.
This warrantee will not cover products
which become defective through misuse
,neglect, or which have been repaired by
other companies.